

Software Engineering Code of Ethics and Professional Practice

Prepared by:

Md. Aquib Azmain

Lecturer

Department of Computer Science & Engineering
BRAC University.

Outline

- Introduction.
- Need of Software Ethics
- The ACM Code of Ethics
- The IEEE Code of Ethics
- The Impact of Unethical Practices
- Ethical Decision-Making Models

INTRODUCTION

- **Ethics** in software engineering is about making decisions that respect user privacy, promote fairness, and benefit society.
- It involves -
 - Adhering to professional codes,
 - Navigating ethical dilemmas, and
 - Understanding the impact of unethical practices.

Need of Software Ethics



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- The quality of software is critical to the society.
- Lives depend on the safety and reliability of many software systems (control of air traffic, nuclear power stations etc).
- The quality of software product depends on the **ethics and professional conduct of the engineers.**

The ACM Code of Ethics

- This code, developed by the **Association for Computing Machinery (ACM)**, serves as a roadmap for making decisions that are ethically responsible and professionally sound.

The ACM Code of Ethics : Moral Imperatives

- Contribute to society and human well-being
- Avoid harm to others
- Be honest and trustworthy

The ACM Code of Ethics : Professional Responsibilities

- Strive for excellence
- Maintain confidentiality
- Respect intellectual property

The ACM Code of Ethics : Professional Leadership Principles

- Lead by example
- Promote public knowledge
- Mentor younger professionals

The IEEE Code of Ethics

- The **Institute of Electrical and Electronics Engineers (IEEE)** has developed this code. Total Eight Principles:
 1. Public
 2. Product
 3. Judgment
 4. Client and Employer
 5. Management
 6. Profession
 7. Colleagues
 8. Self

The IEEE Code of Ethics: Responsibilities to the Public

- Public Safety and Welfare
- Environmental Considerations
- Public Understanding

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The IEEE Code of Ethics: Responsibilities to the Clients and Employers

- Confidentiality
- Conflict of Interest
- Quality Assurance

The IEEE Code of Ethics: Responsibilities to the Profession

- Professional Development
- Professional Conduct
- Support for Colleagues

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www.dilbert.com scottadams@aol.com



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The Impact of Unethical Practices

- Legal Consequences
 - Lawsuits, fines, and in some cases, criminal charges.
 - Collecting user data without proper consent can cause fines under laws like the General Data Protection Regulation (GDPR) in the European Union.
- Reputational Damage
- Impact on Users and Society

Solution: Ethical Decision-Making Models

1. **The Consequence-based model:** This model focuses on the outcomes of an action and aims to choose the action that results in the most good or least harm.
2. **The Duty-based model:** This model emphasizes the inherent duties and obligations of the professional, regardless of the outcome.
3. **The Virtue-based model:** This model focuses on the character and virtues of the individual, aiming to choose actions that align with these virtues.

References

- <https://www.acm.org/code-of-ethics>
- <https://www.ieee.org/about/corporate/governance/p7-8.html>